Windows 11 Optimizer

Safe, reversible tuning for Windows 11 gaming PCs. One button does the job in the right order, and every change can be undone.

User manual, version v4.23

New here?
Open Quick start. Six steps, about 30 minutes, most of it waiting.

Something went wrong?
Open Troubleshooting. Find your symptom and follow the fix.

Want every detail?
The Reference at the back lists each tweak and each tool.

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Before you start

Two minutes of reading that keep you safe.

Read this first

- **Use at your own risk.** These scripts edit the Windows registry and power settings. Use them only on a PC you own. Always keep the restore point the script creates.
- Built for **Windows 11 desktops used for gaming**. On a laptop expect more heat and faster battery drain (processor minimum 100%, power throttling off).
- The gains are small and situational. GPU drivers, in-game settings, XMP/EXPO and cooling matter far more. See the honest notes on each tweak below.
- Some antivirus programs flag `.bat` files that edit the registry. Every file is plain text: open and read it before you run it.

What you need

- A desktop or laptop with **Windows 11** (build 22000 or newer) and an **administrator** account. On older Windows the program warns you first.
- About 30 minutes. Most of it is waiting for the restore point and Windows Update.
- **System Protection** (System Restore) switched on for drive C:, so the restore point can be made. The program tries to switch it on for you. Troubleshooting explains the manual way.
- An internet connection is best. Without one, the online steps are skipped and the rest still works.
- All folders of the download stay together. Do not move a `.bat` file out of its folder.

What it does and does not do

It does	It does not
Make a restore point before it changes anything	Delete your files or games
Apply documented registry and power tweaks, each with its own undo file	Edit game files, inject into games or add overlays
Install recommended drivers from Windows Update, never a maker's graphics driver or firmware	Install third-party programs or "driver updater" tools
Show your PC specs, health and crashes in plain words	Promise FPS numbers: the gains are small and depend on the game
Update itself only after you say yes, with a SHA-256 check	Send your data anywhere: it contacts github.com for updates and Microsoft for drivers

You stay in charge

Easy Setup shows its plan first and asks before it starts. Press **Q** at that prompt to leave without changing anything.

Quick start

Six steps from download to done. Follow them in order.

1 Download the program

- Open **github.com/Kensukeimagery/win11-optimizer** and click the green **Download latest ZIP** button. You need no account.
- Direct link: github.com/Kensukeimagery/win11-optimizer/releases/latest/download/win11-optimizer.zip

You should see a file named `win11-optimizer.zip` (about 1 MB) in your Downloads folder.

2 Unblock the ZIP and extract it

- Right-click the ZIP, choose **Properties**, tick **Unblock** at the bottom and press **OK**. No Unblock box? Skip this. It stops Windows from warning about every file.
- Right-click the ZIP again, choose **Extract All**, then **Extract**. Open the new folder.
- Do not run anything from inside the ZIP.

You should see the files below. Only three of them are for you to open.

0_Easy_Setup (Run as Administrator).bat	Start here
1_Start_Here - PC_Optimizer_Master (Run as Administrator).bat	main menu, for choosing yourself
2_Revert_Everything - System_Restore (Run as Administrator).bat	undo everything
More Tools	extra tools, all reachable from the menu
docs, reg, reg_undo, repair-tools, tools	keep them next to the files above
README.md, CHANGELOG.md, LICENSE, VERSION	information only

3 Start Easy Setup as administrator

- Right-click **0_Easy_Setup (Run as Administrator).bat** and choose **Run as administrator**.
- Windows asks "Do you want to allow this app to make changes?" Press **Yes**.
- A blue "Windows protected your PC" window means SmartScreen does not know the file yet. You downloaded it from the project page yourself, so click **More info**, then **Run anyway**.

You should see a black window titled "Easy Setup".

4 Choose the language and answer one question

- Press **E** for English or **T** for Thai, then Enter. If Thai letters look broken (question marks or boxes), choose E.
- "Do you mainly play games on this PC?" Press **Enter** for No, or **Y** for Yes. A Yes adds the mouse, keyboard, timer and CPU boost tweaks, and opens a second question: **Maximum** applies every optional tweak at once. Turning Memory Integrity off (tweak 13) needs its own typed **YES**, because it lowers protection.
- Not sure? Press Enter. You can add the optional tweaks later (main menu, key 2).
- Read the plan and press **Y** to start.

```
================================================================
   PC OPTIMIZER - EASY SETUP
================================================================
   Do you mainly play games on this PC? Y/N (Enter = N):
   Your answer: not mainly games (the safe settings only).

   What will happen (about 20 to 40 minutes; Windows may ask for one restart):
   1. Check if a newer version of this tool exists (you decide).
   2. Make a safety point (System Restore) so you can always go back.
   3. Remember how the PC is now, to show before and after.
   4. Internet: connected.
   5. Windows Update: we open it and you press install (we guide you).
   6. Find and install the important drivers (network, sound, Bluetooth, chipset).
   7. Restart if Windows needs it, then start this again to finish.
   8. Apply the safe speed settings (the recommended ones only).
   9. Optional: test the internet and pick a faster DNS (skipped unless you say yes).
   10. Check everything again and show a one-page summary.
   Nothing is deleted and every change can be undone. A restore point is made first.
   Start now? Y = start, Q = quit (Enter = start)
```

What the plan looks like. Your list may differ slightly: a laptop or no internet changes lines 4 and 8.

5 Let it work through the ten steps

It prints "Step 3 of 10" and so on. Most steps need nothing from you.

Step	What you see	What you do
1 Update check	"Check for a newer version of this tool now?"	Press Enter. If a newer version exists, you may update.
2 Restore point	"Making the safety point... 1 to 5 minutes"	Wait. Do not close the window.
3 Save the state	"Saved how the PC is now"	Nothing.
4 Internet	"The internet works." or "No internet"	No internet: plug in a cable or connect Wi-Fi and press **R**. Press **S** to skip the online steps. The settings still apply.
5 Windows Update	The Windows Update page opens	Click **Check for updates**, install everything, come back to the black window and press **Enter**. Press **S** to skip this step.
6 Drivers	"Drivers installed: 0. ..."	Nothing. If the graphics card has only the basic driver, it offers the maker's download page: press **Y** to open it.
7 Restart	"Windows needs a restart..." (only when it does)	Press **R**. After the restart, open 0_Easy_Setup again and press Enter at "Continue where it stopped?"
8 Settings	"Applying the settings..." about 1 minute	Nothing.
9 DNS	"Test the internet speed and pick a faster DNS?"	Press Enter (No). The effect is small.
10 Summary	The SUMMARY screen, after 30 seconds to 2 minutes	Read it (next step).

> ⚠ **Do not click inside the black window while it works**
>
> A click selects text and pauses the program. If the title bar starts with **Select**, press **Esc** and it continues.

6 Read the summary and restart

```
================== SUMMARY ==================
   Speed settings in place: 6 of 24 before, 24 of 24 now.
   Devices without a working driver: 0 before, 0 now.
   Drivers installed by this setup: 0.

   Things worth a look (hints, not faults):
   - ...
   To go back: run 2_Revert_Everything, or use Windows System Restore and pick
Before_PC_Optimizer.
   Please restart the PC so every setting takes full effect.
   L = I will restart later, R = restart now (Enter = later)
```

Example only. Your numbers will differ.

- **Speed settings in place** compares the PC before and after. The second number should equal the total.
- **Things worth a look** are hints, not faults. Troubleshooting explains the common ones.
- Press **R** to restart now, or **L** to restart later. The settings take full effect after a restart.
- The full report is saved as `Logs\EasySetup_date-time.txt`.

You are done

After a big Windows update, press **C** in the main menu (`1_Start_Here`) to find settings that Windows reset and put them back.

What Easy Setup does

Every step, and why it comes in this order.

#	Step	What it does	Why at this point
1	Newer version	Asks if you want to check; if a newer version exists you can update this tool.	You start with the latest fixes.
2	Restore point	Creates the System Restore point `Before_PC_Optimizer`.	Everything after it can be undone.
3	Remember now	Counts the speed settings in place and the devices without a driver, and saves the PC specs.	To show before and after.
4	Internet	Checks the connection. If there is none, explains how to fix it (a network driver from the board or laptop maker, copied with a USB drive).	Updates and drivers need it.
5	Windows Update	Opens Windows Update. You press Check for updates, install everything and restart if it asks.	Updates can change settings, so they come first.
6	Drivers	Installs the recommended drivers from Windows Update (network, sound, Bluetooth, chipset). No full driver copy: the restore point covers it. If the graphics card only has the basic driver, opens the maker's download page.	The same safe system as 7_Driver_Check.
7	Restart	Offered only when Windows needs it. Run the file again to continue.	New drivers need it.
8	Safe speed settings	Menu 1, 3 and 4. The gaming answer adds mouse, keyboard, timer resolution and CPU boost. Maximum also adds fullscreen exclusive and the Ndu fix. A laptop skips what makes it hotter or drains the battery.	Last, so updates and drivers cannot undo them.
9	Faster DNS	Optional network test and DNS choice. The default is no.	Small effect, never forced.
10	Check and summary	Settings in place before and after, devices without a driver, things worth a look, how to undo, and a restart reminder.	You see the result on one page.

Not included on purpose

Driver cleanup, deep file cleanup, network reset, changing DNS without asking, removing Windows apps and installing programs. They are for special cases and are in the main menu and repair-tools. Memory Integrity off (13) is applied only when you pick Maximum and type YES.

The PC is kept awake while it runs. Every run is saved as `EasySetup_date-time.txt` in the Logs folder. To see the plan without changing anything, start it from a command window with `-Preview`.

Which button do I press?

Pick the row that matches your situation.

Your situation	Open or press	What happens
New Windows, first time, or not sure	**0_Easy_Setup** (or **E** in the menu)	Everything in the right order, with a plan, a restore point and a summary.
Windows update reset some settings	**C**	Lists what changed and puts it back after you confirm.
I want to choose each optional tweak (09 to 16) myself	**2**, or **5** for the whole set	Asks one by one. Tweak 13 needs a typed YES.
I want every optional tweak without questions	Easy Setup, then Yes to games, then Yes to Maximum	One go. Memory Integrity off (13) still needs its own typed YES.
One tweak feels wrong (for example the mouse)	**7**, then the number	Puts back the Windows default for that one tweak.
Everything feels wrong	**2_Revert_Everything**	Rolls the whole PC back with System Restore. It restarts the PC.
Check or fix drivers	**D**	Scans, lists, and installs only what you choose.
Web pages or the game connection feel slow	**N**	Ping, jitter, packet loss and DNS speed.
See my specs, health or what uses CPU and RAM	**S**, **H**, **R**	One-page reports. They change nothing.
Ask someone for help	**B**	Makes one file with private details replaced. Nothing is sent.

Easy Setup (E) compared with Run Everything (5)

	E: Easy Setup	5: Run Everything
Settings it applies	Tweaks 01-08, base setup, power settings. The games answer adds 10, 11, 15, 16. Maximum adds 09 and 12. Tweak 13 only after a typed YES.	Tweaks 01-08, base setup, power settings, and it asks about 09 to 16 one by one.
Questions	One or two at the start	One per optional tweak
Also does	Update check, Windows Update guide, drivers, DNS choice, before and after summary, continues after a restart	Nothing else
Time	20 to 40 minutes	A few minutes

Both make a restore point named Before_PC_Optimizer. Running either again makes a new one with the same name, and 2_Revert_Everything goes back to the newest.

Undo: three ways back

Use the smallest one that fixes your problem.

A Revert one tweak: menu 7

- Example: the mouse feels odd after file 10. Type 7, then 10.
- Restores the Windows default from `reg_undo`. T removes the network values, P resets all power plans.
- The script remembers the revert; the status check shows it as SKIPPED and will not offer to apply it again.

B Restore your exact previous values: Backup folder

- Every run exports the old keys to `Backup\date-time`.
- Double-click a file there to restore the value your PC really had (more precise than the Windows default).

C Roll back the whole PC: 2_Revert_Everything

- Uses System Restore to return to the `Before_PC_Optimizer` restore point.
- The PC restarts by itself, and programs installed after that point are removed. Serious problems only.

Which restore point does 2_Revert_Everything use?

The newest one named Before_PC_Optimizer. Running Easy Setup or key 5 again makes a new one with the same name, so the older one is no longer the target.

Keep it working

Status check (menu C) and updates (menu U). More Tools\3_Check_Status does the same as menu C.

Status check

1 Run 3_Check_Status as administrator

- It checks every item and prints each line with the current value against the expected one.

2 Answer the fix questions

- Recommended items: Y = re-apply everything missing, S = choose one by one, N = change nothing.
- Optional items are always asked one by one; file 13 still needs YES.
- The old values are backed up to `Backup\date-time_check` before anything is re-applied.

3 Automatic re-check

- After fixing, the script checks the fixed items again to confirm. Everything is saved to `CheckReport_date-time.txt`. Restart once.

Status	Meaning	What the script does
OK	The value is still as the script set it.	Nothing.
CHANGED	A value was changed, for example by a Windows update.	Can re-apply it.
MISSING	The value is not there at all.	Can re-apply it.
NOT CHOSEN	An optional tweak (09-13) you never answered.	Asks: Y apply, N not now, X never ask again.
SKIPPED	You skipped or reverted it.	Does not ask or change it.
UNKNOWN	Cannot be read, or this PC does not have the option.	Information only.

Updating

U Updating: menu U or 6_Check_For_Updates.bat

- Looks for a newer release on GitHub, shows what is new, and installs it only if you press U. N = not now, S = skip this version, L = open the release page.
- Safe by design: the download is checked against a SHA-256 checksum published with the release. The old version is moved to `Backup\update_oldversion_time`, and if anything fails it is put back. Your Backup folder, logs and reports are never touched.
- The first time you open the menu it asks once whether to check automatically at start (it contacts github.com and sends nothing about you). Press A to turn that on or off. Nothing is ever installed without your choice.
- Updating from v4.5 or older needs one manual download of the ZIP. From v4.6 on, the tool updates itself.

Troubleshooting

Find the symptom in the left column.

Symptom	What it means	What to do
A blue "Windows protected your PC" window	SmartScreen does not know the file. Normal for a new `.bat`.	Click **More info**, then **Run anyway**. Next time, unblock the ZIP before you extract it (Quick start, step 2).
Antivirus blocks or deletes a file	Some antivirus programs flag `.bat` files that edit the registry.	Check that you downloaded from the project page and read the file. Never turn antivirus off. If it still blocks, do not use the file.
"Please right-click this file and choose Run as administrator"	You opened it with a normal double-click.	Right-click the file and choose **Run as administrator**. Nothing was changed.
The window freezes. The title bar starts with "Select"	You clicked inside the black window, which pauses it.	Press **Esc**. It continues.
"The argument ... does not exist" or "not recognized" after typing a command	A typo or the wrong folder.	Open the extracted folder, type `cmd` in the address bar, press Enter, and copy the command instead of typing it.
"The safety point could not be made"	System Protection is off for drive C:.	Start menu, type **Create a restore point**, select C:, press **Configure**, choose **Turn on system protection**, press OK. Run Easy Setup again.
"No internet" in step 4	No connection, or the PC lacks a network driver.	Plug in a cable or connect Wi-Fi and press R. With no network driver, download it from your motherboard or laptop maker on another PC and copy it by USB drive. Or press S to skip the online steps.
The Windows Update page did not open	Some PCs block the shortcut.	Open Settings, then Windows Update, by hand. Or press S to skip this step.
Thai letters are spaced out, boxes or question marks	The console window cannot draw Thai well.	Choose **E** (English). Windows Terminal draws Thai better. This Thai manual stays available.
"An earlier Easy Setup was started... Continue?"	You closed it halfway or restarted.	Press Y (or Enter) to continue where it stopped. Press N to start again.
Settings are back to normal after a Windows update	Windows reset some values. This happens.	Press **C** in the main menu and confirm.
The PC is hotter or the fans are louder	Ultimate Performance and CPU boost keep the processor fast.	Undo 16 (menu 7, then 16), or pick Balanced in Power Options. On a laptop, skip the optional tweaks.
The mouse feels odd	Tweak 10 turned off pointer acceleration.	Menu 7, then 10.
A game complains about security or anti-cheat	Anti-cheat rules belong to the game maker. This tool changes Windows settings only.	Open **S** (PC specs) to see Secure Boot and TPM, which some games need. If you turned Memory Integrity off (13), undo it: menu 7, then 13.
Something broke after a driver install	A new driver does not suit the device.	Device Manager, the device, **Roll Back Driver**. Or use System Restore and pick `Before_Driver_Install`.
I want help from someone	The support bundle collects what they need.	Press **B**. Read the file in `Logs`, then attach it to an issue at github.com/Kensukeimagery/win11-optimizer.

Glossary

The words of this manual in plain language.

Word	Meaning
Administrator	A Windows account that may change system settings. Programs that change settings need "Run as administrator".
Restore point	A snapshot of Windows settings that System Restore can return to. It does not touch your documents.
Registry	The database where Windows stores its settings. Each tweak here changes a few values in it.
.reg file	A small text file that sets registry values. You can open it in Notepad and read it. Every tweak has one, and one undo file.
Driver	Software that lets Windows talk to a device such as the graphics card or sound chip.
Windows Update	Microsoft's service for fixes and many drivers.
SmartScreen	A Windows check that warns about files it does not recognise.
SHA-256 checksum	A fingerprint of a file. The updater compares it to the published one to make sure the download is intact.
BIOS / UEFI	The firmware settings screen you reach at power-on (often Delete or F2).
XMP / EXPO	A memory profile in the BIOS that runs the RAM at its rated speed.
Secure Boot, TPM	Security features that Windows 11 and some games, such as Valorant, require. The PC specs page shows whether they are on.
FPS and 1% lows	Frames per second. The 1% lows are the slowest moments, and they decide how smooth a game feels.
Ping, jitter, packet loss	Delay to a server, how much that delay jumps around, and the share of packets that never arrive. Jitter is what you feel as rubber-banding.
DNS	The service that turns a website name into an address. It is used once when you connect and does not change in-game ping.
Refresh rate (Hz)	How many times per second the screen updates. A 144 Hz screen running at 60 Hz looks like input lag.
Memory Integrity	A Windows security feature (HVCI). Turning it off can raise FPS on some PCs but protects the PC less.
Power plan	A set of power rules. Ultimate Performance keeps the processor fast and never sleeps cores.
Core parking	Windows putting idle processor cores to sleep to save power. The power settings here turn it off.
SSD, HDD	Solid-state drive and hard disk drive. Windows and games run faster from an SSD.

Questions and answers

Is it safe?

It makes a restore point first, changes only documented Windows settings, and each tweak has an undo. You can read every file before you run it. Use it only on your own PC, and keep the restore point.

Will it raise my FPS?

Slightly, and only in some games. The tweaks are small. GPU driver settings, in-game settings, XMP/EXPO and cooling matter far more. Be careful with any program that promises large gains.

Will it get me banned in a game?

It does not edit game files, inject code or add overlays. It changes Windows settings only. Anti-cheat rules belong to each game maker, so check yours.

What does it send over the internet?

The update check contacts github.com and sends nothing about you. Driver search uses Windows Update. The network test pings public DNS servers when you start it. The support bundle replaces private details and is never sent.

Can I use it on a laptop?

It is built for desktops. On a laptop Easy Setup skips the tweaks that raise heat and drain the battery.

Can I run it twice?

Yes. Running again is safe. It makes a new restore point with the same name, and 2_Revert_Everything goes back to the newest one.

How do I remove it?

There is no installer. Undo the tweaks with 2_Revert_Everything or menu 7, then delete the folder.

Where are my reports?

In the `Logs` folder next to the program. The newest three of each kind are kept.

Does it cost anything?

No. It is free and open source (MIT license).

Reference

Lookup pages. You do not need them for a first run.

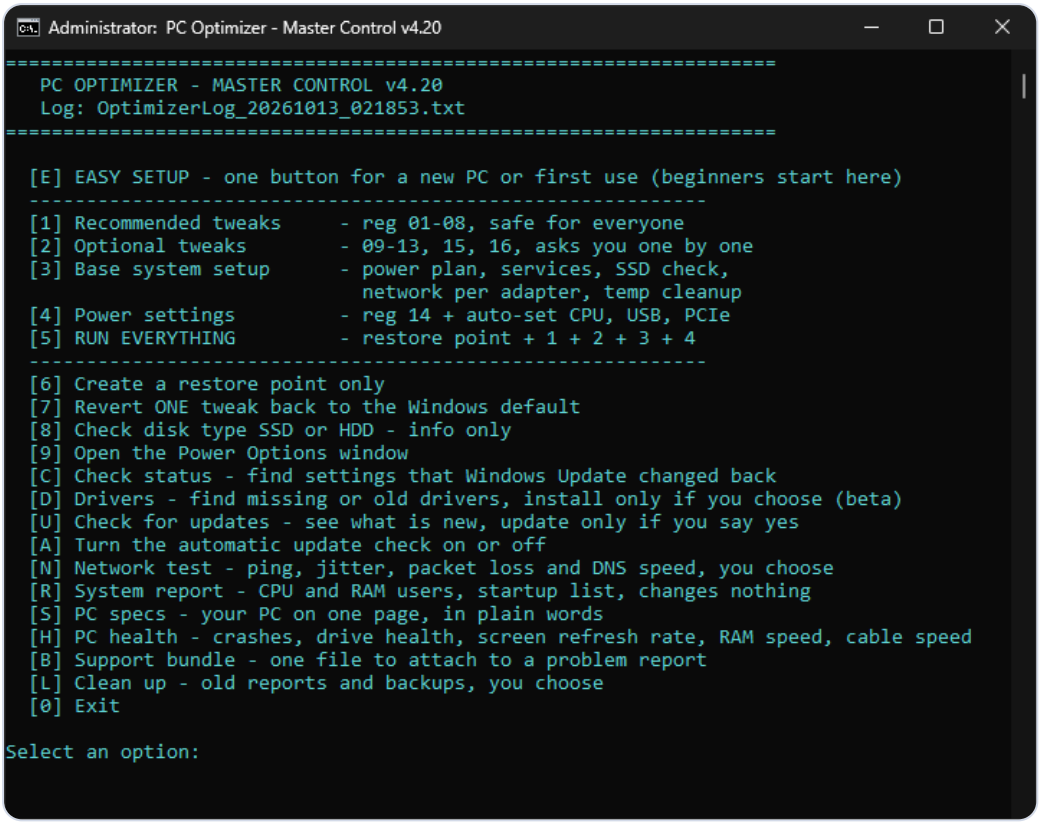
Folder layout

File / folder	Purpose
0_Easy_Setup (Run as Administrator).bat	ONE BUTTON for a new Windows install and for beginners: restore point, drivers, Windows Update (guided), safe settings, summary. Thai or English.
1_Start_Here - PC_Optimizer_Master (Run as Administrator).bat	Main menu. Start here; everything is reachable from it.
2_Revert_Everything - System_Restore (Run as Administrator).bat	Roll the whole PC back with System Restore (serious problems only).
More Tools/	The extra tools below (each one also opens from the main menu). Keep this folder next to the main files.
3_Check_Status (Run as Administrator).bat	Check which tweaks are still applied, for example after a Windows update.
4_Network_Test.bat	Ping, jitter, packet-loss and DNS speed test. Changes nothing unless you choose a DNS at the end. No administrator rights needed to run it.
5_System_Report.bat	Read-only report: what uses your CPU and RAM, what starts with Windows, disk space, GPU driver age. No administrator rights needed.
6_Check_For_Updates.bat	Looks for a newer version on GitHub and installs it only if you say yes. No administrator rights needed.
7_Driver_Check (Run as Administrator).bat	BETA. Finds devices with no driver and lists Windows Update driver updates. Installs only if you choose.
S_PC_Specs.bat, 8_PC_Health.bat, 9_Support_Bundle.bat	Read-only: your PC on one page, a health check (crashes, drives, battery, refresh rate, RAM and cable speed), and ONE file with the reports for a problem report (private details replaced, nothing sent). No administrator rights needed.
tools/	PowerShell helpers used by the menu, the status check, the two tools above, the DNS change and the updater. Do not delete.
reg/1_Recommended	Files 01-08: safe for most PCs.
reg/2_Optional	Files 09-13 and 15: the script asks about each one. (16 is a power setting, see the optional tweaks.)
reg/3_Power	File 14: unlocks hidden CPU options in Power Options.
reg_undo/	One undo file per tweak (Windows defaults), used by menu 7.
repair-tools/	Four stand-alone repair tools. Unrelated to the Master menu.
Backup/ (created when you run)	Copy of every key before it is changed, in a dated folder.

Logs/ (created when you run)	The log of every command and the report of every tool. The newest 3 of each kind are kept and older ones are removed automatically (menu L changes the number).

Main menu keys

Key	What it does
E	Easy Setup: one button for a new PC or first use
1	Recommended tweaks 01-08, safe for everyone
2	Optional tweaks 09-13, 15, 16, asked one by one
3	Base setup: power plan, services, SSD check, network per adapter, temp cleanup
4	Power settings: tweak 14 and the automatic CPU, USB and PCIe settings
5	Run everything: restore point, then 1, 2, 3 and 4
6	Create a restore point only
7	Revert one tweak to the Windows default
8	Show the disk type (SSD or HDD)
9	Open the Power Options window
C	Check status: find settings that Windows Update changed back
D	Drivers: find missing or old drivers (beta)
U	Check for updates; update only if you say yes
A	Turn the automatic update check on or off
N	Network test: ping, jitter, packet loss and DNS speed
R	System report: CPU and RAM users, startup list
S	PC specs on one page, in plain words
H	PC health: crashes, drives, refresh rate, RAM speed, cable speed
B	Support bundle: one file to attach to a problem report
L	Clean up old reports and backups
0	Exit



The main menu (1_Start_Here). E is Easy Setup. Every tool in More Tools has a key here.

Status check: what it checks and its limits

What it checks: all 14 .reg files (read from the same files that are applied), the power plan (Ultimate Performance, or High Performance if Ultimate is unavailable), the four power values, Prefetcher (SSD must be 0, HDD must be 3), TcpAckFrequency and TCPNoDelay on every adapter, and the DiagTrack / dmwappushservice services. Optional 15, 16 and (once used) 17 are checked too (16: boost mode 2 and preference 0 while plugged in).
Limits: it tells you a value is still there, not that it still has an effect (item 08 can show OK on a GPU that does not support GPU scheduling). Values under GameConfigStore (01, 09) are rewritten by Windows and the Xbox Game Bar, so CHANGED there is common. Adding a new VPN or Wi-Fi adapter makes Network show CHANGED until it is configured.

.reg files, group 1: Recommended (01-08)

Menu 1. Meant to be safe for most PCs

01 GameBar_DVR_Off

- Turns off Xbox Game Bar background recording (Game DVR), at user and policy level. Game Mode is kept on (v3 turned it off by mistake).

Helps: gives back CPU, GPU and disk time used for constant recording. No official Microsoft documentation for these values.

02 Power_Throttling_Off

- PowerThrottlingOff = 1 stops Windows slowing background programs. Energy estimation off. Fast Startup off, so Shutdown is a real clean restart.

Helps: Discord or OBS in the background are not slowed down. Cost: slightly longer boot, and more power use on laptops.

03 Gaming_Priority_MMCSS

- NetworkThrottlingIndex = ffffffff removes the network limit for multimedia tasks.
- SystemResponsiveness = 10, the lowest value that takes effect. Microsoft documents that values below 10 are raised to the default of 20, so a value of 0 does nothing.
- Tasks\Games: GPU Priority 8, Priority 6, Scheduling Category High, SFIO Priority High.

Helps: queues work so the game gets CPU, GPU and disk first.

04 CPU_Foreground_Priority

- Win32PrioritySeparation = 0x26: the active window gets longer, higher-priority CPU time slices (default is 0x2).

Helps: the game in front gets more CPU than background programs.

05 Telemetry_Off

- AllowTelemetry = 0, CEIPEnable = 0. On Home and Pro, Windows uses the lowest level it allows; true 0 only applies to Enterprise.

Helps: less background data sent. Tiny effect on FPS; mostly privacy.

06 System_Responsiveness

- Menus appear instantly, frozen apps are closed faster, lighter animations and visual effects.
- NTFS last-access time stamps off; the classic Copy To / Move To right-click entries restored.

Helps: Windows feels quicker, fewer needless disk writes.

07 Delivery_Optimization_Off

- DODownloadMode = 0.

Helps: Windows Update does not upload updates to other PCs while you play online.

08 GPU_Scheduling_On

- HwSchMode = 2 turns on hardware-accelerated GPU scheduling. Needs a restart and a supported GPU and driver; unsupported systems ignore it.

Helps: slightly lower CPU-to-GPU latency. Microsoft says most users should not expect a big change.

Optional tweaks (09-13, 15, 16)

The script asks about each one. Answer for how you actually play

09 Fullscreen_Exclusive

- Forces true Fullscreen Exclusive instead of letting Windows convert games to borderless.
- Answer Y only if fullscreen games stutter or hang on alt-tab. Some newer games work better the default way.

Helps: lower input lag in older games that support Fullscreen Exclusive.

10 Mouse_No_Acceleration

- Turns off Enhance Pointer Precision so the mouse moves 1:1.

Helps: consistent aim in shooters.

11 Keyboard_Fast_Repeat

- KeyboardDelay = 0 (shortest delay before repeat). Flags = 122 stops the Filter Keys pop-up when Shift is held for 8 seconds in a game.

Helps: faster key response and no interrupting pop-up.

12 Ndu_Memory_Fix

- Disables the Ndu service (a memory leak reported on some PCs) and sets DisablePagingExecutive = 1.
- Side effects: Task Manager stops showing per-app network history. Keeping the kernel in RAM costs memory, so skip it on PCs with 8 GB or less.

Helps: only if your RAM use keeps creeping up. The Ndu leak comes from user reports, not Microsoft documentation.

13 Memory_Integrity_Off

- Turns off Memory Integrity (HVCI). You must type YES, and restart.
- Microsoft confirms it can affect gaming performance on some PCs (no figure given). With it off, Windows protects less against kernel-level malware.
- Skip it on a work PC or one holding important data. Undo with menu 7, then 13.

Helps: the only tweak here that can raise FPS noticeably, at a security cost.

15 Timer_Resolution

- Sets GlobalTimerResolutionRequests = 1. Since Windows 10 2004 and Windows 11, a game's timer-resolution request only applies to that game. This value lets such a request apply system-wide again.
- Can steady frame pacing in some games. Microsoft notes that a higher timer resolution uses more power and can keep the CPU from saving power. Needs a restart. Undo: menu 7, then 15.

Helps: frame pacing in some games, effect varies. The value comes from the tweaking community; Microsoft does not document it.

16 CPU_Boost_Aggressive (power setting, no .reg file)

- Plugged in only: processor performance boost mode = Aggressive (2) and energy performance preference = 0 (favour performance), on the active power plan.
- Your earlier values are saved first, so menu 7, then 16 restores exactly those. The Ultimate Performance plan already uses these values on most PCs, so this mainly matters on High Performance or Balanced. Expect more heat, especially on laptops.

Helps: 1% lows in games limited by the CPU. Skipped automatically on PCs that do not offer these settings.

Group 3: Power (14) and automatic settings

Menu 4 does the whole job, no manual setup

14 Unlock_CPU_Power_Settings

- Makes visible the CPU options Windows hides: core parking, maximum frequency, boost mode, energy performance preference.
- Merges files 11 and 12 of v3 and drops entries that are normally visible.

P What menu 4 then sets with powercfg

- Minimum processor state = 100%. Core parking minimum = 100% (cores never sleep).
- USB selective suspend = off. PCIe Link State Power Management = off.
- Plugged-in values only; the battery profile on a laptop is not touched.

Helps: CPU and GPU do not switch to power-saving states mid-game.

What menu 3 (Base setup) does

The parts that are not plain .reg files

Item	What it does	Note
Power plan	Uses Ultimate Performance; High Performance if that is missing; leaves Balanced if neither exists.	The plan GUID is remembered, so repeat runs do not create duplicates (v3 did).
Services	Windows Search and Print Spooler on; DiagTrack and dmwappushservice off.	"already running" in the log is normal.
Prefetcher	Checks the type of the drive Windows is installed on (C:) first: SSD off, HDD on, unknown type left alone.	Stops you disabling it on an HDD, which would slow it down.
Network	Sets TcpAckFrequency and TCPNoDelay on every network adapter.	v3 set these globally, where Windows ignores them. Only affects games using TCP. Microsoft advises testing before changing these.
Old values	Removes MaxUserPort, TcpTimedWaitDelay and misplaced TCP values from v3.	None of them matter for gaming.
Temp files	Deletes the user's temporary files.	Files in use are skipped safely.

Windows settings to change by hand

Things a script cannot do for you

Already automatic in v4, no need to do by hand: GPU scheduling (08), Delivery Optimization (07), USB selective suspend and PCIe power saving (menu 4), Game Bar DVR (01), Game Mode kept on (01), Enhance Pointer Precision (10, if you answered Y).

Clear effect: do these	Where	Why
GPU power = Maximum Performance	NVIDIA Control Panel / AMD Software	Stops the card down-clocking under uneven load.

Network adapter power saving off	Device Manager > adapter > Power Management	Avoids ping spikes.
Add game folders to Defender exclusions	Virus & threat protection > Exclusions	Faster level loading. Does not turn Defender off.
Turn off GPU-software overlays	NVIDIA App / AMD Software > Settings	Less background CPU/GPU load.
Disable audio enhancements	Control Panel > Sound > Enhancements	Lower audio latency.
Cap FPS 3-4 below the refresh rate	In-game limiter, RTSS or NVIDIA Control Panel	Steadier frame times, lower input lag.
NVIDIA Reflex / Low Latency Mode	In-game setting, or NVIDIA Control Panel	Lower input lag when the GPU is the limit.
Shader Cache Size = Unlimited (NVIDIA)	NVIDIA Control Panel > Manage 3D settings	Fewer stutters from shaders being rebuilt.
Use a network cable, not Wi-Fi	Router to PC	The biggest fix for ping spikes and jitter.

Situational or small effect	Where	Why
Focus Assist while gaming	Settings > System > Notifications	No notifications stealing focus.
Xbox Game Bar fully off (if you do not use Win+G)	Settings > Gaming > Xbox Game Bar	One less background process.
Disable fullscreen optimizations per game	Right-click .exe > Properties > Compatibility	For older games with problems.
GPU vendor telemetry services off	services.msc	Less background work.
Move Automatic Maintenance	Control Panel > Security and Maintenance	No scans during games.
Transparency and HDR off if unused	Settings > Personalization / Display	Small saving.
Exclude game folders from Search indexing	Settings > Privacy & security > Searching Windows	Less disk I/O.
Stop unused background apps	Settings > Apps > Installed apps	Frees a little RAM.
Fixed-size page file	System Properties > Advanced > Virtual Memory	More stable over time.

Make sure of: GPU drivers installed from the vendor directly (NVIDIA settings: `GPU_Driver_Guide_EN.pdf`), XMP/EXPO enabled in the BIOS, Variable Refresh Rate on if your monitor supports it, and games set to use the "High performance" GPU on dual-GPU laptops.

Diagnostic tools

These tools look at your PC. They change nothing unless you choose to at the end (a DNS in the network test, drivers in the driver check). Menu N, R, D, S, H and B run the same tools All of them are in the More Tools folder.

4 4_Network_Test.bat

- Sends 30 pings each to your router, 1.1.1.1, 8.8.8.8 and, if you type one, a game server (these are ordinary ping packets). It shows average ping, jitter and packet loss, grades each line Good / Fair / Poor and says where the problem most likely is.
- **Jitter** is how much the delay jumps around. It is what you feel as rubber-banding, and it matters more than a slightly higher average ping.
- The result is saved as `NetworkTest_date-time.txt` in the Logs folder.
- **DNS speed:** it then times real lookups of 8 popular sites on your current DNS and on Cloudflare (1.1.1.1), Google (8.8.8.8) and Quad9 (9.9.9.9), and says how much faster the best one is. After that it asks whether to change DNS. You choose the number, or press Enter to change nothing.

17 DNS servers (optional, you choose)

- Sets the IPv4 DNS servers of your active Ethernet and Wi-Fi adapters to the provider you pick. VPN and virtual adapters are left alone. Windows asks for administrator permission when you confirm.
- Your earlier DNS settings are saved first (including "obtain automatically"). To undo: run the test again and choose U, or use menu 7, then 17.
- Check Status shows this item once you have used it, and re-applies it if a Windows update or a VPN changed it.

How much does it matter? DNS is used once, when you connect to a site or a game server. It does **not** change in-game ping or FPS. A faster DNS can make web pages and launchers start a little quicker, and a broken DNS can make sites fail to load. If your current DNS is within about 5 ms of the best one, leave it alone. The DNS provider you choose sees the names of the sites you visit.

What you see	Most likely cause	What to try
Router line is Fair or Poor	Wi-Fi signal, a bad cable, or other devices using the network.	Use a cable, move closer, pause other downloads and streams.
Router is Good, internet lines are Poor	Your internet provider or the line.	Restart the router; if it stays, contact the provider.
Router and internet Good, only the game server is bad	The route to that server, or the server itself.	Choose another server or region. Nothing on your PC fixes this.
Game server shows "No reply"	Many servers block ping.	Not a problem by itself.

5 5_System_Report.bat

- Lists Windows, CPU, RAM, graphics card and driver age, disks and free space, the power plan, the programs using the most CPU and RAM (browsers with many processes are added together), and the programs that start with Windows.
- Ends with plain hints: long uptime, RAM nearly full, an old GPU driver, a drive with less than 15% free, a failing disk, too many startup programs.
- Saved as `SystemReport_date-time.txt`. Your Windows user name is hidden in the file, but check it before you share it. Program names and paths are still listed.

7 7_Driver_Check (BETA, menu D)

- Scans for devices with no working driver, shows your graphics card and its driver age, and lists the driver updates Windows Update offers. Made mainly for a fresh Windows install, useful any time.
- It always asks first: A = install the recommended ones, E = everything listed, S = choose one by one, G = open the official graphics driver page, C = clean up old driver versions, N = nothing (default).
- Installing uses Windows Update, so the drivers are Microsoft-signed. Before the first install it creates a restore point and saves a copy of your current drivers in `Backup\drivers_time`. It installs one at a time, never restarts the PC itself, and tells you how to go back: Roll Back Driver in Device Manager, System Restore, or the saved copy.
- **Not installed by this tool:** the graphics driver from the maker, BIOS and firmware. For the graphics card it opens the official NVIDIA, AMD or Intel page. For the full chipset package, BIOS and Wi-Fi/LAN it names your motherboard or laptop so you can find its support page. Windows Update graphics drivers are skipped when you already have the maker driver, because they can be older. The small chipset INF packages that Windows Update offers (named like "Intel chipset driver package 10.1.1.38") are listed as optional.
- Low-impact items (for example processor sub-devices and monitors) are listed as optional, grouped by maker, and are not part of "recommended".
- On a fresh Windows with no network driver, Windows Update cannot be reached. Install the LAN or Wi-Fi driver from your motherboard or laptop maker first (use another PC and a USB drive).
- After installing it asks Windows Update again and tells you what is left. An item that "failed" because an earlier package already covers the device is counted as "not needed", not as an error.
- **C = clean up (beta).** Lists (A) old versions of drivers you still use, which Windows keeps in its driver store (for example older NVIDIA versions, often several hundred MB each), and (B) entries of devices that are no longer connected (old USB devices, monitors). The newest version, any version in use and boot-critical packages are always kept. Before removing anything it creates a restore point `Before_Driver_Cleanup` and saves a copy of each package in `Backup\drivers_removed_time`; if a copy cannot be saved, that package is left alone. Nothing is forced: if Windows says a driver is still needed, it stays. Network, storage and system entries are listed but never removed, and Bluetooth entries are kept because the device may only be switched off. Packages from Microsoft, and groups where the highest version number is not also the newest date (the maker changed its numbering), are left alone and named in the list. Driver packages for hardware that is only unplugged right now are deliberately left alone.
- Honest expectation for C: it frees disk space and tidies Device Manager. It does not raise FPS or lower ping.
- Beta: installing drivers has not been tested on many PCs. Read the list before you answer, and keep the restore point.

S S_PC_Specs.bat (menu S)

- For people who do not know where to find their specs: one page, plain words, no jargon needed. It shows the computer and motherboard, Windows, the processor (cores and threads), memory (size, type DDR4/DDR5, speed, slots used), graphics card (video memory and driver), every screen with its resolution and refresh rate, every drive (SSD or hard drive, size, free space, which one holds Windows), the network link, and whether Secure Boot and the TPM are on (Windows 11 and some games need them).
- **Check-up at the end:** do the drivers and settings fit together? It checks that every device has a working driver, that the graphics driver comes from the maker, which network, graphics, sound or Bluetooth drivers are older than 4 years, whether the optimizer settings are still in place (Windows updates can reset them), whether each screen runs at its highest refresh rate, whether the RAM runs at its rated speed and whether the network cable runs at full speed.
- A short note under some parts says what the number means (for example how much RAM is usually enough). They are general guidelines, not verdicts.
- No serial numbers, MAC addresses or IP addresses are shown, so the page can be shared when you ask for help or sell the PC. It is saved as `PCSpecs_date-time.txt` and is also included in the support bundle.

8 8_PC_Health.bat (menu H)

- Read-only. It looks at: (A) blue screens, unexpected shutdowns, graphics driver resets and programs that crashed in the last 30 days (one that crashes often is shown with its file location), with the blue screen code explained in plain words; (B) drive health, temperature and wear when the drive reports them; (C) battery wear on laptops; (D) things that matter for games: whether each screen runs at its highest refresh rate, whether the RAM runs at its rated speed (XMP / EXPO), and whether the network cable link is slower than the adapter supports.
- Items worth a look are marked CHECK. They are hints, not faults: one blue screen is not proof of a broken part. A screen that supports 144 Hz but runs at 60 Hz can feel like input lag, and no tweak can fix that.
- Administrator rights are not needed; drive temperature and wear appear more often when you run it as administrator. The result is saved as `PCHealth_date-time.txt`.

L L_Clean_Up.bat (menu L)

- **Reports and logs** go to the `Logs` folder. The newest 3 of each kind are kept (a support bundle keeps 5) and older ones are removed automatically when the menu or a tool runs. Only files with exactly the names this program creates are ever removed. A menu session in which nothing was run leaves no log at all. Reports saved next to the scripts by older versions are moved into Logs.
- **Backups** in the `Backup` folder are never removed by themselves. Menu L lists them with their size and, only after you say yes, removes the older ones: it keeps the newest full copy of your drivers, the newest 2 driver-cleanup copies, the newest 2 earlier versions of this program and the newest 10 registry copies.
- Menu L also lets you change how many reports are kept (0 = never remove reports).

9 9_Support_Bundle.bat (menu B)

- Collects the version, Windows build, Check Status, PC health, system report, a driver scan (this PC only, nothing is installed), your saved choices and the latest logs into ONE file, `SupportBundle_date-time.txt`. Attach that file to a problem report.
- Private details are replaced: your user name, the computer name, home network addresses (192.168..., 172.16-31..., 169.254...), hardware (MAC) addresses and e-mail addresses. The file still lists program names and your PC model, so read it before you share it.
- Nothing is changed and nothing is sent anywhere. The file stays in the folder; you decide whether to share it.

What really helps steady FPS and low ping

- Cap the frame rate 3-4 below your monitor's refresh rate (in the game, RTSS or the NVIDIA Control Panel). It gives steadier frame times than an uncapped game that pins the GPU.
- Turn on NVIDIA Reflex (or Low Latency Mode) if your game offers it.
- Use a network cable. Wi-Fi jitter is the most common cause of bad ping.
- Keep GPU drivers current, keep the PC cool and close heavy programs before you play.
- None of the registry tweaks in this project lowers ping by a noticeable amount, and neither does changing DNS. They are small, situational gains.

Folder: repair-tools

Targeted repair tools. Not needed on every run. Right-click > Run as administrator

A Deep_Clean_Junk_Files.bat

- Runs Disk Cleanup on the safe categories plus temp files. It **never** touches your Downloads folder, and it asks before emptying the Recycle Bin or deleting old Windows setup files such as Windows.old. Reports free space before and after, and warns if less than 15% is left.

Use when: disk space is low, or every 1-2 months.

B Network_Reset_Fix_Internet.bat

- Resets Winsock, IPv4 and IPv6, renews the IP address, flushes DNS and ARP, then pings to test.

Use when: the connection drops, will not connect, or is oddly slow.

C Bluetooth_Fix.bat

- Re-enables the pairing service and restarts the Bluetooth Support Service.

Use when: a device will not pair or connect.

D System_File_Scan_Repair.bat

- DISM first, then SFC, the order Microsoft recommends (15-40 minutes). ResetBase is an optional Y/N question because it cannot be undone.

Use when: programs crash, freeze often, or you suspect damaged system files.

External tools (optional)

Not part of this repository. Use only if you want them

Read before use

Both tools download and run the latest code from the internet each time. What was checked in October 2026 may differ from a future version. Create a restore point first (menu 6) and run 3_Check_Status afterwards to see which of this project's values changed.

Tool	FPS effect	Real benefit	Risk	For
WinUtil: Install tab	None	Installs several programs at once through winget	Low	Everyone
WinUtil: Tweaks tab	Almost none	Overlaps almost entirely with this project	Medium: may change values set here	Not recommended after this project
RemoveWindowsAI	Very little to none	Removes Copilot and background AI processes; saves some RAM and disk; privacy	High: force-removes packages Windows protects	People who never use AI, or PCs with 8 GB RAM or less

W Chris Titus Tech WinUtil

- Use the Install tab. Skip the Tweaks tab; if you used it, run 3_Check_Status and re-apply what changed.
- To open: right-click Start > Terminal (Admin), then paste the command from the official GitHub page (Stable):

```
irm https://christitus.com/win | iex
```

Link: github.com/ChrisTitusTech/winutil

R RemoveWindowsAI (zoicware)

- Removes Copilot, Recall and other AI features, and installs packages so Windows Update does not bring them back.
- If you only want Copilot to stop using RAM, you do not need this: turn it off in Settings > Apps > Startup, or uninstall the Copilot app in Settings > Apps > Installed apps.
- Needs **Windows PowerShell 5.1** (search "Windows PowerShell", run as administrator). Its GitHub page says PowerShell 7 has problems.
- Always use Backup Mode. It creates a restore point and is what makes a full revert possible.
- The GitHub page says some antivirus software may complain and suggests turning antivirus off temporarily. **This manual advises against that.** If it is blocked, do not use it.
- Read the Documentation page first: the latest version has options touching Windows Defender's AI components.

Backup Mode (removes everything):

```
& ([scriptblock]::Create((irm "https://raw.githubusercontent.com/zoicware/RemoveWindowsAI/main/RemoveWindowsAi.ps1"))) -nonInteractive -backupMode -AllOptions
```

Revert everything:

```
& ([scriptblock]::Create((irm "https://raw.githubusercontent.com/zoicware/RemoveWindowsAI/main/RemoveWindowsAi.ps1"))) -nonInteractive -revertMode -AllOptions
```

Each command must be on one line. If copying from this PDF splits it, join it before pressing Enter, or copy from the GitHub page. Link: github.com/zoicware/RemoveWindowsAI. Use the full links above, never shortened ones, because you cannot tell where a short link leads.

References

Links are clickable in the PDF. Last checked 30 Sep 2026

Used for	Source / link
03 SystemResponsiveness, Tasks\Games	Microsoft Learn - Multimedia Class Scheduler Service learn.microsoft.com/en-us/windows/win32/procthread/multimedia-class-scheduler-service Values below 10 are raised to 20; a value of 100 turns MMCSS off.
04 Win32PrioritySeparation	Microsoft TechNet - Win32PrioritySeparation technet.microsoft.com/library/Cc976120 Default is 0x2, as in UNDO_04.
07 Delivery Optimization	Microsoft Learn - Delivery Optimization reference learn.microsoft.com/en-us/windows/deployment/do/waas-delivery-optimization-reference
08 GPU scheduling	Microsoft DirectX Blog - Hardware Accelerated GPU Scheduling devblogs.microsoft.com/directx/hardware-accelerated-gpu-scheduling/ Most users should not expect a big change.

13 Memory Integrity	Microsoft Support - Options to optimize gaming performance in Windows 11 support.microsoft.com/en-us/windows/options-to-optimize-gaming-performance-in-windows-11-a255f612-2949-4373-a566-ff6f3f474613 Confirms an effect on some PCs and warns of higher exposure to threats.
Menu 3: TCP per adapter	Microsoft Support KB328890 - TcpAckFrequency support.microsoft.com/kb/328890 Set per interface GUID; Microsoft advises against changing defaults without testing.
Menu 4, power plans	Microsoft Learn - Powercfg command-line options learn.microsoft.com/windows-hardware/design/device-experiences/powercfg-command-line-options
Menus 5, 6: restore point	Microsoft Learn - Checkpoint-Computer learn.microsoft.com/powershell/module/microsoft.powershell.management/checkpoint-computer Normally one restore point per day, so the script sets SystemRestorePointCreationFrequency = 0 first.
15 Timer resolution	Microsoft Learn - timeBeginPeriod function learn.microsoft.com/en-us/windows/win32/api/timeapi/nf-timeapi-timebeginperiod Describes the per-process change since Windows 10 2004 and Windows 11, and the extra power use of a higher resolution.
16 CPU boost	Microsoft Learn - PERFBOOSTMODE and processor power management options learn.microsoft.com/en-us/windows-hardware/customize/power-settings/options-for-perf-state-engine-perfboostmode learn.microsoft.com/en-us/windows-hardware/customize/power-settings/configure-processor-power-management-options Boost mode 2 = Aggressive. Energy performance preference runs 0-100 %.
WinUtil	GitHub - ChrisTitusTech/winutil (README) github.com/ChrisTitusTech/winutil
RemoveWindowsAI	GitHub - zoicware/RemoveWindowsAI (README) github.com/zoicware/RemoveWindowsAI
repair-tools: SFC / DISM	Microsoft Support - Use the System File Checker tool support.microsoft.com/help/929833 Run DISM before SFC, the order used here.

Without official documentation

- 01 GameBar/DVR and 09 Fullscreen Exclusive (GameDVR_FSE* values): shared by the tweaking community; no Microsoft page explains them.
- 02 PowerThrottlingOff, EnergyEstimationEnabled: no public Microsoft documentation.
- 15 GlobalTimerResolutionRequests: reported by the community. The Microsoft timeBeginPeriod page describes the behaviour change but not this value.
- 06 Control Panel\Desktop timeouts, MinAnimate: widely known, not sourced here.
- 11 Keyboard Response flags and 12 Ndu memory leak: from user reports, not Microsoft.
- The defaults in reg_undo for these are typical Windows values. If you need your exact originals, use the Backup folder.

Results differ between PCs and Microsoft may change how these values behave in a later Windows release. Every item can be reverted with menu 7 or the Backup folder.